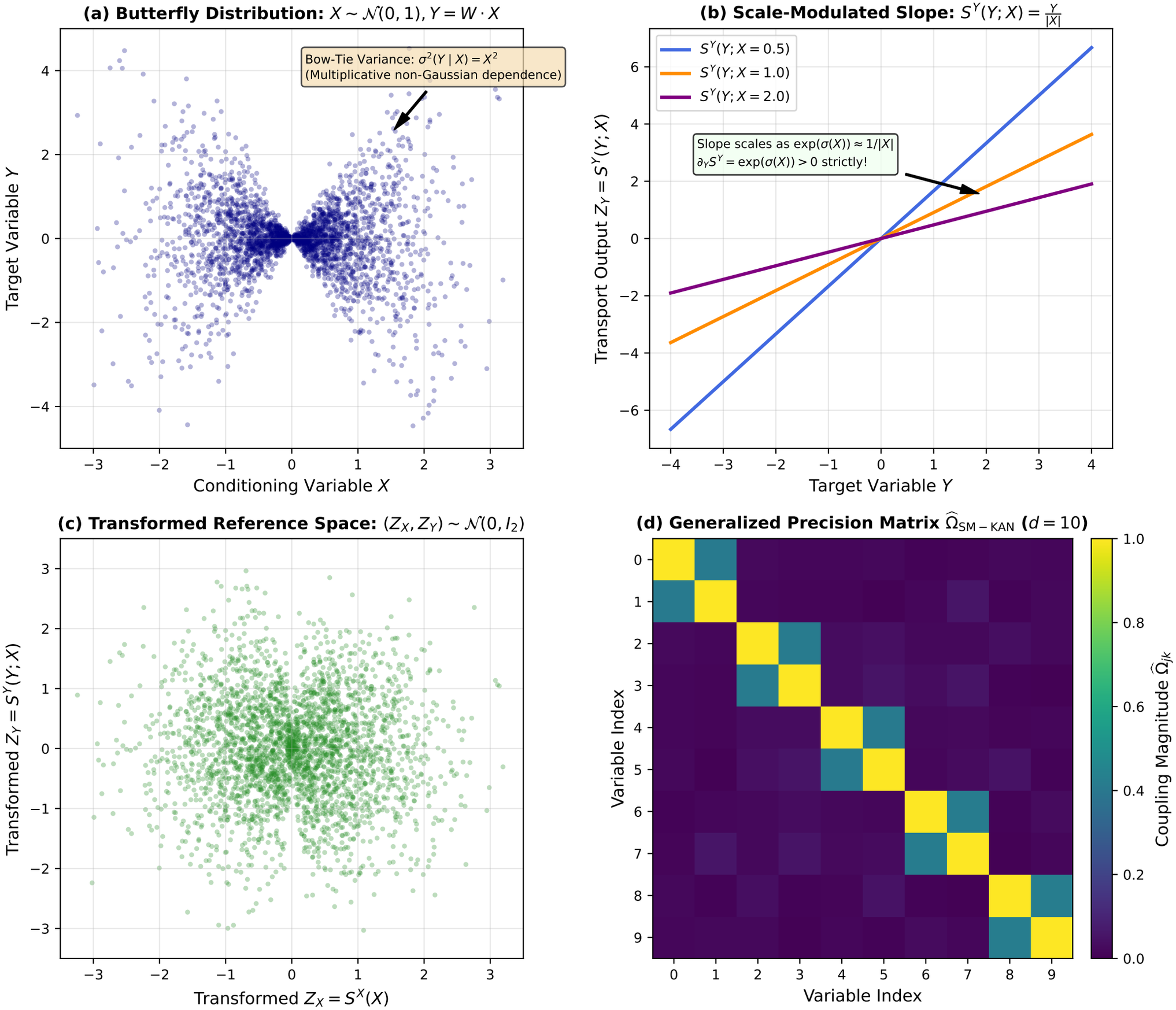

(a) Butterfly Distribution: $X \sim \mathcal{N}(0,1), Y = W \cdot X$
Bow-Tie Variance: $\sigma^2(Y \mid X) = X^2$
(Multiplicative non-Gaussian dependence)
Target Variable $Y$
Conditioning Variable $X$
(b) Scale-Modulated Slope: $S^Y(Y;X) = \frac{Y}{|X|}$
$S^Y(Y;X=0.5)$
$S^Y(Y;X=1.0)$
$S^Y(Y;X=2.0)$
Slope scales as $\exp(\sigma(X)) \approx 1/|X|$
$\partial_Y S^Y = \exp(\sigma(X)) > 0$ strictly!
Transport Output $Z_Y = S^Y(Y;X)$
Target Variable $Y$
(c) Transformed Reference Space: $(Z_X, Z_Y) \sim \mathcal{N}(0, I_2)$
Transformed $Z_Y = S^Y(Y;X)$
Transformed $Z_X = S^X(X)$
(d) Generalized Precision Matrix $\widehat{\Omega}_{\mathrm{SM-KAN}}$ ($d = 10$)
Variable Index
Variable Index
Coupling Magnitude $\widehat{\Omega}_{jk}$